THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 8 · SET 03

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — **1 questions · 12 marks**

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) A perfect square can never end with the digit:

(a) 1 (b) 4 (c) 8 (d) 9

Ans: (c) Squares end only in 0, 1, 4, 5, 6 or 9.

(ii) The cube root of 0.027 is:

(a) 0.03 (b) 0.3 (c) 3 (d) 0.9

Ans: (b) $0.027 = 27/1000 = (3/10)^3 \Rightarrow 0.3$.

(iii) The sum of the first 12 odd natural numbers is:

(a) 24 (b) 144 (c) 132 (d) 156

Ans: (b) Sum of first n odd numbers = $n^2 = 12^2 = 144$.

(iv) The value of $(-3)^{-2}$ is:

(a) 1/9 (b) −1/9 (c) −9 (d) 9

Ans: (a) $(-3)^{-2} = 1/(-3)^2 = 1/9$.

(v) 1999 written in Roman numerals is:

(a) MCMXCIX (b) MDCCCCLXXXXVIIII

(c) MCMXCVIIII (d) MXMIX

Ans: (a) 1000 = M, 900 = CM, 90 = XC, 9 = IX.

(vi) Which of the following is NOT a valid base-4 numeral?

(a) 1302 (b) 2031 (c) 3210 (d) 1234

Ans: (d) Digits in base 4 can only be 0, 1, 2, 3; '4' is not allowed.

(vii) Three angles of a quadrilateral are 80°, 100° and 70°. The fourth angle is:

(a) 110° (b) 100° (c) 90° (d) 120°

Ans: (a) 360° − (80 + 100 + 70) = 110°.

(viii) The diagonals of a rhombus:

(a) are always equal (b) are parallel

(c) do not intersect (d) bisect each other at right angles

Ans: (d) Property of a rhombus: diagonals are perpendicular bisectors of each other.

(ix) Which of the following numbers is divisible by 11?

(a) 2728 + 1 = 2729 (b) 3026

(c) 3025 (d) 3035

Ans: (c) (3 + 2) − (0 + 5) = 0 ⇒ 3025 is divisible by 11 (3025 = 11 × 275).

(x) (x + 2)(x + 3) equals:

(a) $x^2 + 5x + 6$ (b) $x^2 + 6x + 5$ (c) $x^2 + 5x + 5$ (d) $x^2 + 6$

Ans: (a) $x^2 + 3x + 2x + 6$.

(xi) If $a + b = 10$ and $ab = 21$, then $(a - b)^2$ equals:

(a) 79 (b) 58 (c) 100 (d) 16

Ans: (d) $(a - b)^2 = (a + b)^2 - 4ab = 100 - 84 = 16$.

(xii) A car travels 150 km in 3 hours. At the same speed it will travel in 5 hours:

(a) 200 km (b) 300 km (c) 250 km (d) 350 km

Ans: (c) Speed = 50 km/h ⇒ 5 × 50 = 250 km.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the cube root of 1728 by prime factorisation.

$1728 = 2^6 \times 3^3$ (1 mark)
$\sqrt[3]{1728} = 2^2 \times 3 = 12$ (1 mark)

3. Write 3.56×10^5 and 2.1×10^{-3} in usual form.

$3.56 \times 10^5 = 356000$ (1 mark)
$2.1 \times 10^{-3} = 0.0021$ (1 mark)

4. Write 1999 in Roman numerals.

1000 + 900 + 90 + 9 (1 mark) = MCMXCIX (1 mark)

5. Add $(1101)_2$ and $(1011)_2$ and give the answer in base 2.

13 + 11 = 24 (1 mark); $24 = (11000)_2$ (1 mark). Direct binary addition is also accepted.

OR

Convert the decimal number 100 into base 7.

$100 = 2\times49 + 0\times7 + 2$ (1 mark) ⇒ $(202)_7$ (1 mark)

6. In parallelogram ABCD, $\angle A = (2x + 10)°$ and $\angle C = (3x - 20)°$. Find x and $\angle A$.

Opposite angles are equal: $2x + 10 = 3x - 20 \Rightarrow x = 30$ (1 mark)
$\angle A = 70°$ (1 mark)

7. Find the ratio of 3 m 60 cm to 2.4 m in simplest form.

3 m 60 cm = 360 cm; 2.4 m = 240 cm (1 mark)
360 : 240 = 3 : 2 (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Without actual division, check whether 678432 is divisible by (i) 8 (ii) 11. Give reasons.

(i) Last three digits 432 = 8 × 54 ⇒ divisible by 8 (1½ marks)
(ii) (6 + 8 + 3) − (7 + 4 + 2) = 17 − 13 = 4, not 0 or a multiple of 11 ⇒ not divisible by 11 (1½ marks)

9. In parallelogram ABCD, AB = (2x + 3) cm and DC = (3x − 5) cm. Find x and the length of AB.

Opposite sides are equal: $2x + 3 = 3x - 5$ (1 mark) ⇒ $x = 8$ (1 mark)
AB = 2(8) + 3 = 19 cm (1 mark)

OR

The diagonals PR and QS of a rectangle PQRS meet at O. If $\angle POQ = 110°$, find $\angle OPQ$.

Diagonals of a rectangle are equal and bisect each other ⇒ OP = OQ (1 mark)
In ΔOPQ: ∠OPQ = ∠OQP (1 mark) = (180° − 110°)/2 = 35° (1 mark)

10. Using suitable identities, find: (i) 97×103 (ii) 205^2 (iii) 99^2

(i) $(100 - 3)(100 + 3) = 10000 - 9 = 9991$ (1 mark)
(ii) $(200 + 5)^2 = 40000 + 2000 + 25 = 42025$ (1 mark)
(iii) $(100 - 1)^2 = 10000 - 200 + 1 = 9801$ (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 3528 must be divided so that the quotient is a perfect square. Also find the square root of the quotient.
(ii) Find the square root of 0.0289.

(i) $3528 = 2^3 \times 3^2 \times 7^2$ (1 mark); divide by 2 (1 mark); quotient $1764 = 2^2 \times 3^2 \times 7^2$, $\sqrt{} = 42$ (1 mark)
(ii) $0.0289 = 289/10000 \Rightarrow \sqrt{} = 17/100 = 0.17$ (2 marks)

OR

(i) Find the smallest number by which 1125 must be multiplied so that the product is a perfect cube. Also find the cube root of the product.
(ii) The volume of a cubical tank is 0.343 m^3. Find the length of its edge.

(i) $1125 = 3^2 \times 5^3$ (1 mark); multiply by 3 (1 mark); product $3375 = 3^3 \times 5^3$, cube root 15 (1 mark)
(ii) $0.343 = 343/1000 = (7/10)^3 \Rightarrow$ edge = 0.7 m (2 marks)

12. Write in standard form: (i) 4,56,00,000 (ii) 0.00000072. Write in usual form: (iii) 7.3×10^{-4} (iv) 5.2×10^6. (v) Find $2 \times 10^5 + 3 \times 10^4$ in standard form.

(i) 4.56×10^7 (ii) 7.2×10^{-7} (iii) 0.00073 (iv) 5200000 (v) 2.3×10^5 (1 mark each)

OR

Simplify: (i) $(2^3)^{-2} \times 2^7$ (ii) $6^5 \div 6^{-2}$ (iii) $(4^{-1} + 2^{-1})^{-1}$ (iv) $(10^{-3})^{-2}$ (v) $(a^2b)^3 \div (ab)^2$

(i) $2^{-6} \times 2^7 = 2$ (ii) 6^7 (iii) $(1/4 + 1/2)^{-1} = (3/4)^{-1} = 4/3$ (iv) 10^6 (v) $a^6b^3 \div a^2b^2 = a^4b$ (1 mark each)

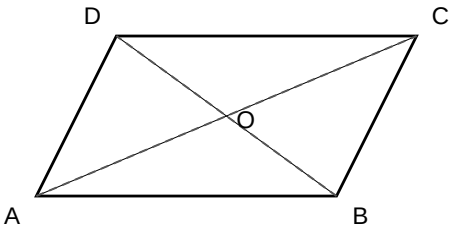
13. (i) Write the ratio of 2.5 kg to 750 g in simplest form.
(ii) Divide ₹ 1,800 among A, B and C in the ratio 2 : 3 : 4.
(iii) If 8 pens cost ₹ 96, find the cost of 15 pens.
(iv) Find x if 6 : x :: 9 : 15.
(v) The ratio of boys to girls in a class is 5 : 4 and there are 36 students. Find the number of girls.

(i) 2500 g : 750 g = 10 : 3 (1 mark)
(ii) Total parts 9, 1 part = ₹ 200 ⇒ ₹ 400, ₹ 600, ₹ 800 (1 mark)
(iii) 1 pen = ₹ 12 ⇒ 15 pens = ₹ 180 (1 mark)
(iv) $9x = 6 \times 15 \Rightarrow x = 10$ (1 mark)
(v) Girls = $4/9 \times 36 = 16$ (1 mark)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A flooring contractor cuts parallelogram-shaped tiles ABCD. In one tile, AB = 12 cm, BC = 8 cm and ∠A = 65°. The diagonals AC and BD intersect at O.



(i) Write the length of CD.

> Opposite sides of a parallelogram are equal: CD = AB = 12 cm

(ii) Find ∠B.

> Adjacent angles are supplementary: ∠B = 180° − 65° = 115°

(iii) Show that AO = OC.

> In ΔAOB and ΔCOD: AB = CD, ∠OAB = ∠OCD, ∠OBA = ∠ODC (alternate angles) (1 mark)
> ⇒ ΔAOB ≅ ΔCOD (ASA), so AO = OC (1 mark)

OR

(iii) Find the perimeter of the tile and the cost of an edging strip around 50 such tiles at ₹ 2 per cm.

> Perimeter = 2(12 + 8) = 40 cm (1 mark); cost = 50 × 40 × 2 = ₹ 4000 (1 mark)

15. Kabir arranges square tiles to form bigger and bigger squares: 1 tile (1×1), 4 tiles (2×2), 9 tiles (3×3), and so on. He notices that each new square needs the next odd number of extra tiles.

(i) How many tiles are needed for the 8th square?

> $8^2 = 64$ tiles

(ii) How many extra tiles are needed to change an 8 × 8 square into a 9 × 9 square?

> 81 − 64 = 17 tiles (the 9th odd number)

(iii) Show that the sum of the first 10 odd numbers is a perfect square and name it.

> 1 + 3 + 5 + … + 19 = 100 (1 mark) = 10^2, a perfect square (1 mark)

OR

(iii) A square arrangement uses 441 tiles. How many more tiles are needed to make the next larger square?

> √441 = 21 (1 mark); next square $22^2 = 484$ ⇒ 484 − 441 = 43 tiles (1 mark)

16. Three friends — Ria, Sana and Tej — travel by a shared taxi for different distances: Ria 3 km, Sana 5 km and Tej 7 km. The total fare of ₹ 450 is divided in the ratio of the distances travelled.

(i) Write the ratio of the shares of Ria, Sana and Tej.

> 3 : 5 : 7

(ii) Find Sana's share of the fare.

> Total parts = 15; 1 part = ₹ 30; Sana pays 5 × 30 = ₹ 150

(iii) Find the difference between Tej's share and Ria's share.

> Tej = 7 × 30 = ₹ 210; Ria = 3 × 30 = ₹ 90 (1 mark); difference = ₹ 120 (1 mark)

OR

(iii) Another day the same three friends pay ₹ 600 for the same distances. Find the amount Ria pays and show that the ratio 3 : 5 : 7 is unchanged.

> 1 part = ₹ 40, Ria pays ₹ 120 (1 mark); shares 120 : 200 : 280 = 3 : 5 : 7 (1 mark)